Week 3: Forward Kinematics and DH Convention

EE0849 Introduction to Robotics - Detailed Notes

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1 Introduction to Forward Kinematics

1.1 The Forward Kinematics Problem

Forward kinematics is one of the fundamental problems in robotics. Given the joint variables (angles for revolute joints, displacements for prismatic joints), we want to determine the position and orientation of the end-effector.

Problem Statement:

- **Given:** Joint variables $q = [\theta_1, \theta_2, \dots, \theta_n]^T$
- **Find:** End-effector pose T_n^0 (position and orientation)

Mathematically, forward kinematics is a mapping:

$$T_n^0 = f(q)$$

This mapping is always unique—for a given set of joint angles, there is exactly one end-effector pose.

1.2 Why Forward Kinematics Matters

Forward kinematics is essential for:

1. **Simulation:** Predicting where the robot will move before commanding it
2. **Control:** Verifying the robot reached the commanded position
3. **Path Planning:** Converting joint-space trajectories to Cartesian-space paths
4. **Visualization:** Rendering the robot in simulators and digital twins
5. **Collision Detection:** Determining link positions to check for obstacles

1.3 Serial Manipulators

Most industrial robots are **serial manipulators**—a chain of rigid links connected by joints. Each joint connects two consecutive links:

- **Link 0** (Base): Fixed to the ground/mounting surface
- **Link n** (End-effector): Carries the tool or gripper

The total transformation from base to end-effector is the product of individual link transformations:

$$T_n^0 = T_1^0 \cdot T_2^1 \cdot T_3^2 \cdots T_n^{n-1}$$

2 Joint Types

2.1 Revolute Joints

A **revolute joint** allows rotation about a single axis. It is the most common joint type in industrial robots.

- **Variable:** θ (joint angle)
- **Fixed:** d (link offset along the rotation axis)
- **Symbol:** or R
- **Example:** Human elbow, shoulder, wrist

2.2 Prismatic Joints

A **prismatic joint** allows linear translation along a single axis.

- **Variable:** d (joint displacement)
- **Fixed:** θ (orientation about the translation axis)
- **Symbol:** or P
- **Example:** Linear actuators, telescoping mechanisms

2.3 Robot Configuration Notation

Robots are often described by their joint sequence:

- **RRR:** 3 revolute joints (e.g., planar arm)
- **RRRRRR:** 6 revolute joints (most industrial robots)
- **SCARA:** RRP configuration (Selective Compliance Assembly Robot Arm)
- **Cartesian:** PPP configuration (gantry robots)

3 The Denavit-Hartenberg Convention

3.1 Historical Background

The DH convention was developed by Jacques Denavit and Richard Hartenberg in 1955. It provides a systematic method for:

1. Assigning coordinate frames to each link
2. Describing the geometric relationship between consecutive frames
3. Using only 4 parameters per link (minimal representation)

3.2 The Four DH Parameters

Parameter	Symbol	Description
Link Length	a_i	Distance along x_i from z_{i-1} to z_i
Link Twist	α_i	Angle about x_i from z_{i-1} to z_i
Link Offset	d_i	Distance along z_{i-1} from x_{i-1} to x_i
Joint Angle	θ_i	Angle about z_{i-1} from x_{i-1} to x_i

Key insight: For a revolute joint, θ_i is the variable and d_i is constant. For a prismatic joint, d_i is the variable and θ_i is constant.

3.3 The DH Transformation Matrix

The transformation from frame $\{i-1\}$ to frame $\{i\}$ consists of four operations:

$$T_i^{i-1} = R_z(\theta_i) \cdot T_z(d_i) \cdot T_x(a_i) \cdot R_x(\alpha_i)$$

Where:

1. $R_z(\theta_i)$: Rotate about z_{i-1} by angle θ_i
2. $T_z(d_i)$: Translate along z_{i-1} by distance d_i
3. $T_x(a_i)$: Translate along x_i by distance a_i
4. $R_x(\alpha_i)$: Rotate about x_i by angle α_i

Multiplying these four matrices gives:

$$T_i^{i-1} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This is the **standard DH transformation matrix**. Memorize it or know how to derive it!

3.4 Frame Assignment Rules

The DH convention requires specific placement of coordinate frames:

3.4.1 Step-by-Step Procedure

1. **Identify joint axes:** Locate all joint axes (rotation axis for revolute, translation axis for prismatic)
2. **Assign z_i axes:** Place z_i along the axis of joint $i+1$
3. **Locate frame origins:**
   - If z_{i-1} and z_i intersect: origin at the intersection
   - If z_{i-1} and z_i are parallel: origin anywhere on z_i
   - If z_{i-1} and z_i are skew: origin at the intersection of z_i and the common normal
4. **Assign x_i axes:** Along the common normal from z_{i-1} to z_i
5. **Assign y_i axes:** Complete the right-hand coordinate system

3.4.2 Special Cases

Intersecting axes ($a_i = 0$): When z_{i-1} and z_i intersect, $a_i = 0$ and x_i is perpendicular to both axes. This is common in spherical wrists.

Parallel axes ($\alpha_i = 0$ or $180°$): When z_{i-1} and z_i are parallel, α_i is either $0°$ or $180°$. This is common in planar robots.

3.4.3 Base and End-Effector Frames

- **Frame 0 (Base):** Usually at the robot base with z_0 along the first joint axis
- **Frame n (End-effector):** At the tool center point (TCP), may require an additional tool transform

4 Worked Examples

4.1 Example 1: 2-Link Planar Arm

Consider a planar arm with two revolute joints and link lengths L_1 and L_2.

4.1.1 DH Parameter Table

Link	θ_i	d_i	a_i	α_i
1	θ_1	0	L_1	0
2	θ_2	0	L_2	0

Note: All z axes are parallel (out of page), so all $\alpha_i = 0$ and all $d_i = 0$.

4.1.2 Forward Kinematics Derivation

Link 1 transformation:

$$T_1^0 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & L_1\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & L_1\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 2 transformation:

$$T_2^1 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & L_2\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & L_2\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Total transformation:

$$T_2^0 = T_1^0 \cdot T_2^1 = \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 & L_1\cos\theta_1 + L_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 & L_1\sin\theta_1 + L_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4.1.3 End-Effector Position

From the transformation matrix, the end-effector position is:

$$x = L_1\cos\theta_1 + L_2\cos(\theta_1+\theta_2)$$

$$y = L_1\sin\theta_1 + L_2\sin(\theta_1+\theta_2)$$

$$z = 0$$

4.2 Example 2: 3-Link Anthropomorphic Arm

A 3-DOF arm with a vertical waist joint and two horizontal joints (shoulder and elbow).

4.2.1 DH Parameter Table

Link	θ_i	d_i	a_i	α_i
1	θ_1	d_1	0	90°
2	θ_2	0	a_2	0
3	θ_3	0	a_3	0

The $\alpha_1 = 90°$ accounts for the perpendicular relationship between the vertical waist axis and the horizontal shoulder axis.

4.3 Example 3: 6-DOF Industrial Robot (FANUC LR Mate 200iD)

4.3.1 Robot Structure

The FANUC LR Mate 200iD is a typical 6-DOF industrial robot:

- **Joints 1-3 (Arm):** Position the wrist center in space
- **Joints 4-6 (Wrist):** Orient the end-effector

4.3.2 DH Parameter Table

Link	θ	d (mm)	a (mm)	α
1	θ_1	330	75	$-90°$
2	θ_2	0	300	$0°$
3	θ_3	0	75	$-90°$
4	θ_4	320	0	$90°$
5	θ_5	0	0	$-90°$
6	θ_6	80	0	$0°$

4.3.3 Spherical Wrist

Joints 4, 5, and 6 form a **spherical wrist**—their axes intersect at a single point (the wrist center). This allows:

1. **Kinematic decoupling:** Position and orientation problems can be solved separately
2. **Dexterous manipulation:** Any orientation can be achieved at the wrist center

5 Python Implementation

5.1 Using roboticstoolbox-python

The `roboticstoolbox-python` library provides convenient tools for robot kinematics.

5.1.1 Installation

```
pip install roboticstoolbox-python
```

5.1.2 Creating a DH Robot

```
from roboticstoolbox import DHRobot, RevoluteDH
import numpy as np

# 2-link planar arm
L1, L2 = 1.0, 0.5

robot = DHRobot([
    RevoluteDH(d=0, a=L1, alpha=0),
    RevoluteDH(d=0, a=L2, alpha=0)
], name='2-Link Planar')

print(robot)
```

5.1.3 Computing Forward Kinematics

```
# Set joint angles (radians)
q = [np.pi/4, np.pi/4]  # 45°, 45°

# Compute FK
T = robot.fkine(q)

print("End-effector transform:")
print(T)

print(f"Position: {T.t}")  # Translation vector
print(f"Rotation:\n{T.R}")  # Rotation matrix
```

5.1.4 Visualization

```
# Plot robot at configuration q
robot.plot(q, block=True)

# Interactive teaching pendant
robot.teach(q)
```

5.2 FANUC Model in Python

```
from roboticstoolbox import DHRobot, RevoluteDH
import numpy as np

# FANUC LR Mate 200iD (dimensions in meters)
fanuc = DHRobot([
    RevoluteDH(d=0.330, a=0.075, alpha=-np.pi/2),
    RevoluteDH(d=0, a=0.300, alpha=0),
    RevoluteDH(d=0, a=0.075, alpha=-np.pi/2),
    RevoluteDH(d=0.320, a=0, alpha=np.pi/2),
    RevoluteDH(d=0, a=0, alpha=-np.pi/2),
    RevoluteDH(d=0.080, a=0, alpha=0),
], name='FANUC LR Mate 200iD')

# Home configuration
q_home = [0, 0, 0, 0, 0, 0]

# Compute FK
T = fanuc.fkine(q_home)
print(f"Home position: x={T.t[0]:.3f}, y={T.t[1]:.3f}, z={T.t[2]:.3f}")
```

6 DH Convention Variants

6.1 Standard vs. Modified DH

There are two common DH conventions:

Aspect	Standard DH	Modified DH (Craig)
Transform order	$R_z T_z T_x R_x$	$R_x T_x R_z T_z$
Frame location	Distal end of link	Proximal end of link

Aspect	Standard DH	Modified DH (Craig)
Common usage	Spong textbook	Craig textbook

Important: This course uses the **Standard DH convention**. When using external resources or libraries, verify which convention is being used!

6.2 Identifying the Convention

To identify which convention a source uses:

1. Check the transformation order in the DH matrix
2. Look at where frames are placed on links
3. Check the textbook or documentation reference

7 Common Mistakes to Avoid

1. **Wrong α sign:** Always follow the right-hand rule for rotation direction
2. **Degrees vs. radians:** Python trigonometric functions use radians!

```
np.sin(90)      # Wrong! This is 90 radians
np.sin(np.pi/2) # Correct: 90 degrees in radians
```

3. **Wrong z axis assignment:** Remember z_i is along joint $i+1$, not joint i
4. **Forgetting the base frame:** Frame 0 must be defined at the robot base
5. **Mixing conventions:** Don't mix Standard and Modified DH parameters

8 Summary

8.1 Key Equations

DH Transformation Matrix:

$$T_i^{i-1} = \begin{bmatrix} c_\theta & -s_\theta c_\alpha & s_\theta s_\alpha & a \cdot c_\theta \\ s_\theta & c_\theta c_\alpha & -c_\theta s_\alpha & a \cdot s_\theta \\ 0 & s_\alpha & c_\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Total transformation:

$$T_n^0 = T_1^0 \cdot T_2^1 \cdots T_n^{n-1}$$

2-Link planar arm position:

$$x = L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

8.2 Key Concepts

1. Forward kinematics maps joint angles to end-effector pose
2. DH convention uses 4 parameters per link: θ, d, a, α
3. For revolute joints: θ is variable, d is constant
4. For prismatic joints: d is variable, θ is constant
5. Frame assignment follows specific rules for z and x axes

9 Practice Problems

9.1 Problem 1

Derive the forward kinematics for a 3-link planar arm with link lengths $L_1 = 1.0$ m, $L_2 = 0.8$ m, $L_3 = 0.5$ m.

9.2 Problem 2

For the FANUC robot, calculate the end-effector position when: - $\theta_1 = 45°$, all other joints at 0°

9.3 Problem 3

A robot has the following DH parameters:

Link	θ	d	a	α
1	θ_1	0.5	0	90°
2	θ_2	0	0.3	0

Compute T_2^0 symbolically, then evaluate at $\theta_1 = 30°$, $\theta_2 = 45°$.